

Quantitative Reasoning-I

Assignment 1 – Solution

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Business Scenario

A student starts a small coffee stall near a university. The monthly fixed cost is Rs. 30,000. Preparing one cup costs Rs. 120 and one cup is sold for Rs. 250. The stall sells 400 cups in a month. A survey shows that 80% of customers are satisfied. Some customers also say that the coffee is good, but the service is slow.

Let x = number of cups sold.

Q1. Quantitative and Qualitative Information

a. Two quantitative examples are:

1. Monthly fixed cost = Rs. 30,000
2. Cost of preparing one cup = Rs. 120

b. One qualitative example is:

Some customers say, "The coffee is good, but the service is slow."

c. Difference:

Quantitative information is in numbers and can be measured, like 30,000, 120, 250, 400 and 80%. Qualitative information describes something in words, like good coffee or slow service.

Q2. Variables, Constants and Coefficients

- a.** x represents the number of cups sold.
- b.** One constant is 120, which is the cost of preparing one cup.
- c.** In $120x$, 120 is the cost per cup.

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Q3. Expressions and Basic Calculations

a. Total Cost when x cups are sold:

$$TC = \text{Fixed Cost} + (\text{Cost per cup} \times x)$$

$$TC = 30,000 + 120x$$

b. Total Revenue when x cups are sold:

$$TR = \text{Selling price per cup} \times x$$

$$TR = 250x$$

c. When 400 cups are sold:

$$TC = 30,000 + 120(400)$$

$$TC = 30,000 + 48,000 = \mathbf{78,000}$$

$$TR = 250(400) = \mathbf{100,000}$$

Q4. Calculation and Interpretation

a. Total Cost:

$$TC = 30,000 + 120(400)$$

$$TC = 30,000 + 48,000 = \mathbf{Rs. 78,000}$$

b. Total Revenue:

$$TR = 250(400)$$

$$TR = \mathbf{Rs. 100,000}$$

c. Amount left after subtracting the cost from revenue:

$$\text{Profit} = TR - TC$$

$$\text{Profit} = 100,000 - 78,000$$

$$\text{Profit} = \mathbf{Rs. 22,000}$$

d. Meaning of the result:

The coffee stall makes Rs. 22,000 profit in one month after paying its fixed and variable costs. So, at 400 cups, the business is earning money.

Final Answers

- Total Cost for 400 cups = Rs. 78,000
- Total Revenue for 400 cups = Rs. 100,000
- Profit = Rs. 22,000
- 80% of customers are satisfied, although some mention slow service.

Date Submitted: **28 Sep 2026** Instructor Remarks: _____